
Silent Drain: Unveiling Smartphone Battery Depletion Through Usage and Electro-Thermal Dynamics

Summary

Smartphone battery endurance is jointly affected by internal battery characteristics and user usage behavior. To quantitatively analyze battery depletion under realistic conditions, this paper establishes a system-level modeling framework integrating battery dynamics with user scenarios.

For task 1, an **electro-thermal battery model** is established to describe the discharge behavior of a smartphone battery. The model captures the **coupled evolution of state of charge, terminal voltage, and battery temperature** under dynamic power loads, and successfully reproduces key discharge phenomena such as voltage drop under high load and temperature-dependent performance degradation.

For task 2, a **Time-to-Empty (TTE) prediction model** is developed based on the electro-thermal coupled dynamics. Battery depletion is defined as the first occurrence of **SOC exhaustion or terminal voltage reaching a cutoff threshold**, allowing the model to capture early shutdown caused by voltage sag under high load. A time-stepping numerical scheme is implemented to compute TTE under different usage scenarios, producing scenario-dependent battery life estimates and explaining the impact of high-power activities on battery endurance.

For task 3, **sensitivity analysis** is conducted based on the proposed models to identify the **key factors affecting battery life** and to assess model robustness. The results indicate that **user power demand and effective battery capacity explain most of the variation** in battery endurance, while thermal-related parameters have a smaller but non-negligible impact. The analysis also demonstrates that the model remains stable under reasonable parameter perturbations.

For task 4, based on the modeling and analysis results, **practical recommendations** are provided for users to improve battery endurance. The results suggest that **reducing sustained high-power usage** and avoiding prolonged operation under unfavorable thermal conditions can effectively extend battery life. For example, reducing screen brightness from 100% to 50% can increase TTE by approximately **20%–35%** across gaming, video, and browsing scenarios, while reducing CPU/GPU load during heavy gaming can extend battery life by about **15%–25%**.

Finally, this study provides a **clear and interpretable framework** for understanding smartphone battery depletion by integrating physical battery modeling with user behavior analysis. The proposed approach offers **quantitative insights** into battery performance and supports battery-aware usage strategies.

Keywords: Smartphone Battery Depletion; Electro-Thermal Model; Sensitivity Analysis; Time-to-Empty; Battery Life Prediction

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1 Introduction

1.1 Problem Background

Smartphones have become indispensable in modern daily life, making battery life one of the most critical factors influencing user experience. Despite significant advances in battery technology, users often observe substantial variability in battery discharge behavior: even under similar initial charge levels and comparable usage durations, smartphones may exhibit markedly different discharge rates. This inconsistency poses challenges for both users and system designers, as unpredictable battery drain can severely limit device reliability and usability.

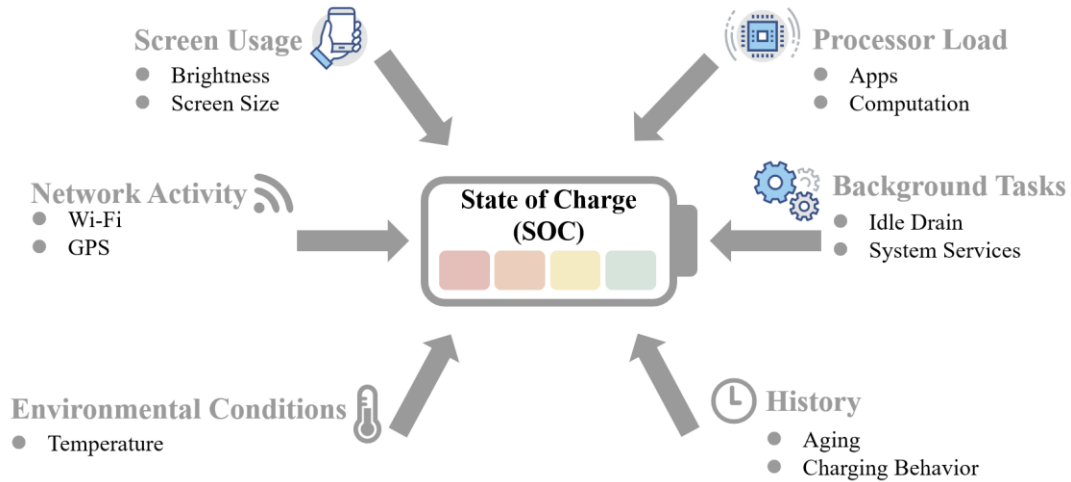


Figure 1: Key Contributors to Smartphone Battery Drain

Such variability arises from the complex interplay of multiple factors, including screen usage, processor workload, network connectivity, background applications, and environmental conditions. These external operating states interact with the internal electrochemical dynamics of lithium-ion batteries, leading to highly nonlinear and time-varying discharge behavior. Traditional empirical rules or discrete models are insufficient to capture these continuous interactions or to explain rapid battery depletion under realistic usage scenarios. Since battery discharge is inherently a continuous and dynamic process, a continuous-time mathematical model provides a more natural and effective framework for describing the evolution of the state of charge, predicting battery depletion time, and identifying the dominant drivers of power consumption.

1.2 Restatement of the Problem

The objective of this problem is to model and analyze the battery performance of a smartphone under realistic usage conditions. Specifically, we are required to:

- Construct a continuous-time mathematical model to describe the evolution of the battery's state of charge (SOC) during smartphone operation;
- Incorporate key factors such as user power demand, environmental temperature, and battery characteristics into the model;
- Determine the time-to-empty of the battery under different usage scenarios;
- Identify which factors contribute most significantly to battery depletion and which have relatively minor effects;
- Evaluate the robustness and limitations of the proposed model and provide insights

that may guide practical power-management strategies.

1.3 Our Work

We construct two main models to analyze smartphone battery depletion. The first model describes the electro-thermal behavior of the battery under dynamic power loads, while the second model characterizes user usage patterns and their impact on battery consumption.

Based on these models, we simulate the discharge process and evaluate battery performance under different usage scenarios. A sensitivity analysis is further conducted to identify the key factors influencing battery life and assess model robustness.

Finally, based on the modeling results, we provide practical suggestions for users to improve battery endurance.

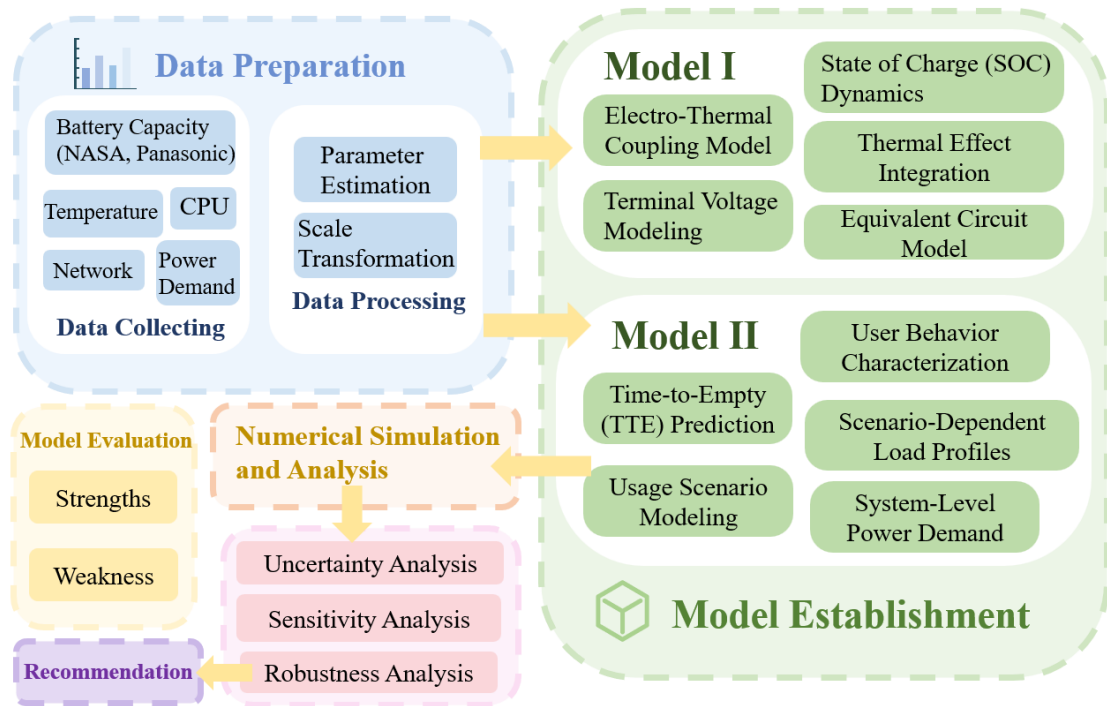


Figure 2: The Process of Our Work

2 Assumptions

- **Assumption 1:** The study focuses exclusively on the discharge process. Charging events and external power supplies are not considered during the simulation window.

Justification: The primary objective is to characterize battery depletion and predict time-to-empty (TTE). Charging involves distinct electrochemical dynamics and control strategies that are outside the scope of depletion analysis.

- **Assumption 2:** The battery is treated as a lumped-parameter system, neglecting spatial variations in current density and temperature gradients.

Justification: Smartphone batteries are small and protected by thermal management systems. A lumped model sufficiently captures the dominant system-level coupling between temperature, internal resistance, and capacity without the computational cost of spatial simulation.

- **Assumption 3:** The battery's electrical behavior is described by a first-order

Thevenin equivalent circuit, where the terminal voltage is determined by the Open-Circuit Voltage (OCV) and polarization effects.

Justification: This model effectively balances complexity and accuracy, capturing key transient phenomena like voltage sag under load and recovery during idle periods, which are critical for predicting voltage-limited shutdowns.

- **Assumption 4:** The total power consumption is modeled as the additive sum of major functional components (screen, CPU, network, etc.), and power conversion efficiency is assumed constant for a given mode.

Justification: At the system level, interaction effects between hardware modules are secondary to their individual power demands. This modular approach allows for the construction of realistic, scenario-based usage profiles.

- **Assumption 5:** Battery depletion is defined as the first occurrence of either the SOC reaching 0% or the terminal voltage dropping below the cutoff threshold (V_{cutoff}).

Justification: In high-power scenarios, voltage sag may trigger a system shutdown before the charge is fully exhausted. Considering both criteria is essential for accurately predicting runtime under heavy loads.

- **Assumption 6:** In the absence of proprietary smartphone battery data, model parameters are calibrated using the Panasonic 18650PF dataset (for electro-thermal properties) and NASA B0005 dataset (for aging trends), assuming similar lithium-ion chemistry behaviors.

Justification: Both datasets reflect fundamental lithium-ion mechanisms. Using valid scaling factors allows these datasets to serve as reliable proxies for analyzing temperature sensitivity and aging in smartphone batteries

3 Notations

The key mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Definition
$SOC(t)$	State of charge of the battery
$I(t)$	Battery discharge current
$V_{term}(t)$	Battery terminal voltage
$V_1(t)$	Polarization voltage
$T(t)$	Battery temperature
$OCV(SOC)$	Open-circuit voltage
$R_0(T)$	Ohmic internal resistance
$C_{avail}(T, Age)$	Effective available capacity
V_{cutoff}	Cut-off voltage for shutdown

***Note:** Here is a partial notation. Detailed explanations for each symbol can be found in the corresponding text.

4 Model I: Electrothermal Coupled Dynamic Model

To capture the dynamic and premature battery depletion observed in smartphones, we developed an electro-thermal coupled dynamic model that links user-driven power demand with the internal electrical and thermal states of a lithium-ion battery. Rather than modeling the state of charge (SOC) in isolation, the proposed framework integrates interacting components.

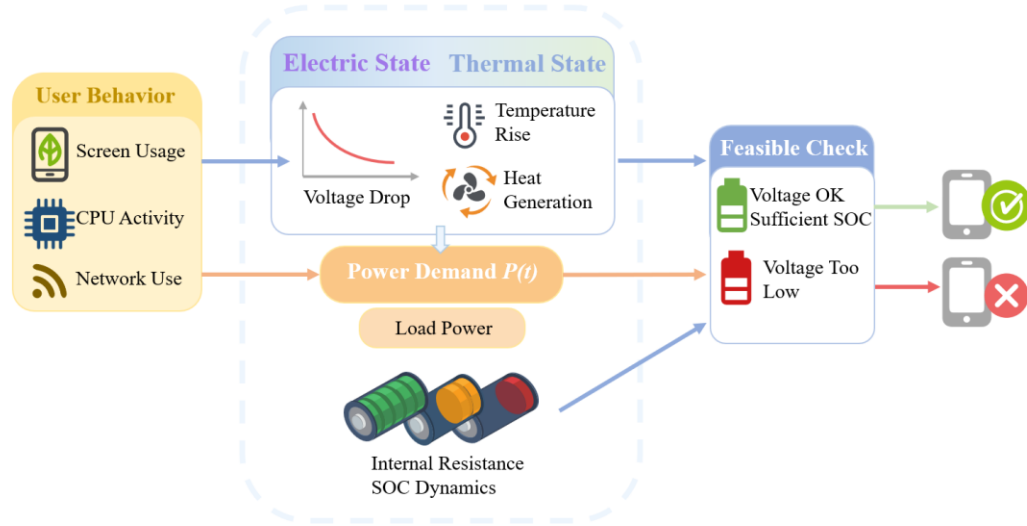


Figure 3: Electro-Thermal Coupled Battery Model

4.1 Modeling Components and Structure

4.1.1 SOC-Dynamic equation

$$\frac{d SOC(t)}{dt} = -\frac{I(t)}{C_{avail}}. \quad (1)$$

4.1.2 Electrical Model and Terminal Voltage

The electrical behavior of the battery is described using a first-order Thevenin equivalent circuit. The model consists of an open-circuit voltage source $OCV(SOC, T)$, a series ohmic resistance $R_0(T)$, and a polarization branch composed of a resistance $R_1(T)$ and a capacitance C_1 . The voltage across the polarization branch is denoted by $V_1(t)$.

Under the discharge convention $I(t) > 0$, the terminal voltage is given by

$$V_{term}(t) = OCV(SOC(t), T(t)) - I(t)R_0(T(t)) - V_1(t). \quad (2)$$

This voltage determines whether the battery can sustain the required load power.

4.1.3 Polarization Dynamics

The polarization voltage $V_1(t)$ captures transient electrochemical effects associated with charge redistribution. Applying Kirchhoff's current law to the polarization branch yields

$$I(t) = \frac{V_1(t)}{R_1(T(t))} + C_1 \frac{dV_1(t)}{dt}, \quad (3)$$

which can be rearranged as

$$\frac{dV_1(t)}{dt} = \frac{1}{C_1} I(t) - \frac{1}{R_1(T(t))C_1} V_1(t). \quad (4)$$

When the load decreases, $V_1(t)$ decays exponentially, leading to partial recovery of the terminal voltage.

4.1.4 Thermal Dynamics

Battery discharge generates heat, which in turn affects internal resistance and capacity. The battery temperature is modeled using a lumped thermal balance,

$$mc_p \frac{dT(t)}{dt} = \dot{Q}_{\text{batt}}(t) + \lambda P_{\text{cpu}}(t) - hA(T(t) - T_{\text{env}}), \quad (5)$$

where mc_p is the thermal capacitance of the battery, hA represents heat dissipation to the environment, and λ denotes the fraction of processor power conducted to the battery.

Under the first-order circuit model, internal heat generation is expressed as

$$\dot{Q}_{\text{batt}}(t) = I(t)^2 R_0(T(t)) + \frac{V_1(t)^2}{R_1(T(t))} \quad (6)$$

4.1.5 Load Power Characterization

$$P_{\text{load}}(t) = P_{\text{base}} + P_{\text{screen}}(t) + P_{\text{cpu}}(t) + P_{\text{net}}(t) + P_{\text{gps}}(t) \quad (7)$$

4.1.6 Power–Current Coupling and Feasibility

The battery must supply the required load power subject to its instantaneous electrical state. This constraint is expressed by the power balance

$$P_{\text{load}}(t) = \eta V_{\text{term}}(t) I(t), \quad (8)$$

where η is the power conversion efficiency.

Substituting the terminal voltage expression yields an implicit equation for the discharge current. Introducing an effective resistance $R_{\text{tot}}(T) = R_0(T) + R_1(T)$, the current can be obtained from the physically admissible root of the resulting quadratic equation,

$$I(t) = \frac{(OCV - V_1) - \sqrt{(OCV - V_1)^2 - 4R_{\text{tot}}P_{\text{load}}(t)}}{2R_{\text{tot}}}. \quad (9)$$

If the discriminant becomes negative, the demanded power cannot be delivered by the battery under its current internal state, indicating a brownout or shutdown event.

4.2 Load Power Characterization and Usage Scenarios

To apply the electro-thermal coupled model under realistic operating conditions, the user-driven load power must be specified for representative smartphone usage scenarios.

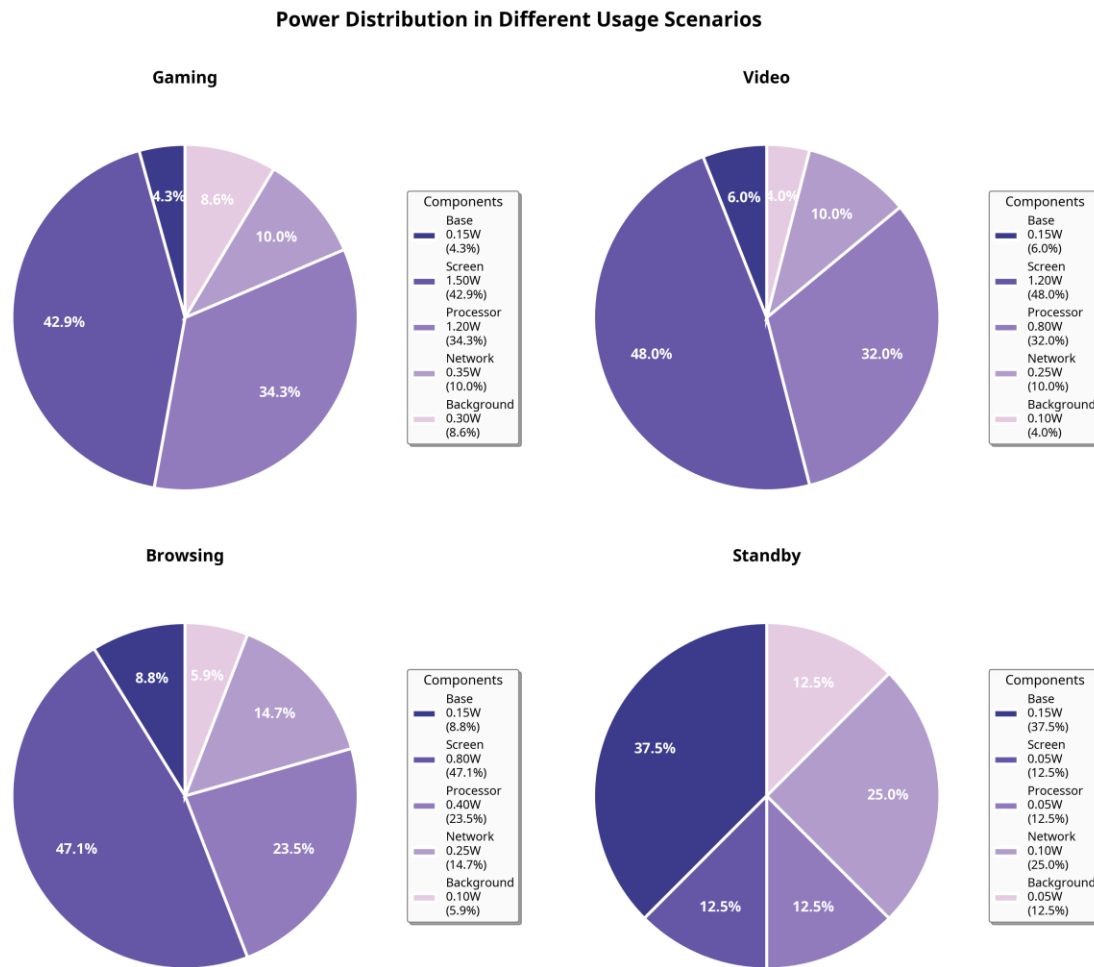


Figure 4: Power Distribution in Different Usage Scenarios

Figure 4 shows the relative contributions of baseline system power, screen, processor, network, and background activity in gaming, video, web browsing, and standby modes. The results highlight substantial differences in power structure across usage scenarios, providing the basis for the scenario-dependent load power model used in the battery discharge simulation.

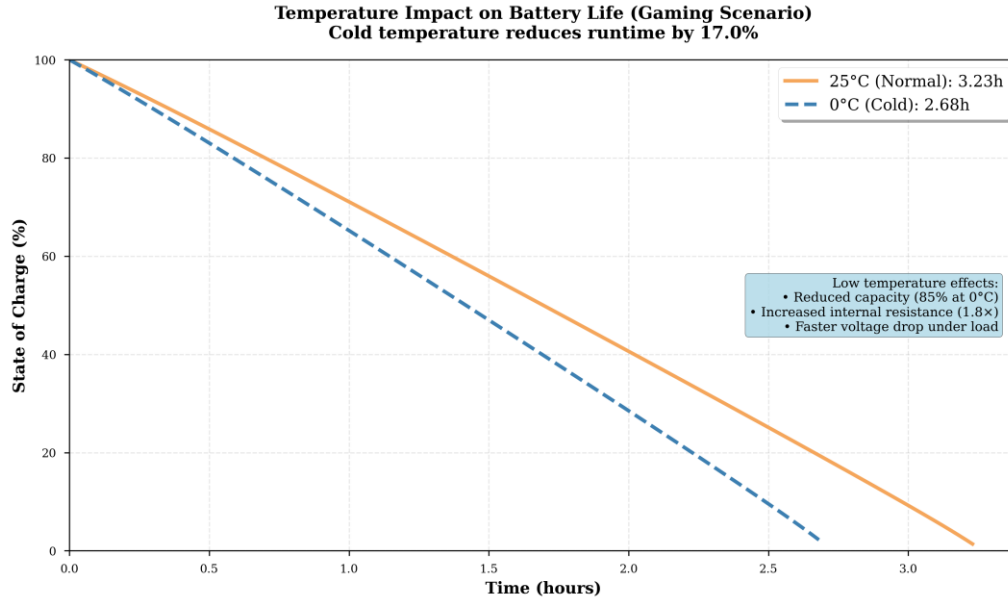


Figure 5: Temperature Impact on Battery Life (Gaming Scenario)

The Figure 5 compares the SOC evolution at normal temperature (25 °C) and cold temperature (0 °C). Under cold conditions, the predicted time-to-empty is reduced by approximately 17%, reflecting the combined effects of reduced effective capacity, increased internal resistance, and accelerated voltage drop under high load. These results demonstrate the strong sensitivity of battery runtime to temperature in high-power usage scenarios.

4.3 Parameter Estimation and Calibration

We estimate parameters for Electrothermal Coupled Dynamic Model using two open datasets: (i) Panasonic 18650PF HPPC tests to identify the first-order Thevenin electrical parameters and their temperature dependence, and (ii) NASA B0005 cycling data to quantify aging trends. Since these laboratory cells differ from a smartphone battery in form factor, we apply a capacity-based remapping to obtain a phone-level parameter set.

4.3.1 Data Acquisition and Preprocessing

We perform three preprocessing steps to ensure data reliability:

- (1) merge duplicated timestamps;
- (2) unify current sign convention;
- (3) detect rest intervals ($|I| < \epsilon$) for OCV extraction and relaxation fitting.

4.3.2 Electrical parameters from HPPC

OCV–SOC. For each rest plateau, the open-circuit voltage is approximated by the terminal voltage near the end of the rest period and then interpolated to obtain a continuous OCV(SOC).

Ohmic resistance R_0 . For a current pulse of amplitude I_p , we compute the instantaneous voltage drop and estimate

$$R_0^{ref}(\text{SOC}) \approx \frac{V_{pre} - V_0^+}{I_p} \quad (10)$$

then aggregate multiple valid pulses (e.g., median) within each SOC bin.

Polarization branch R_1, C_1 . Under a constant pulse current I_p , the first-order Thevenin model predicts an exponential transient. We fit the pulse and relaxation voltage using

$$V_{term}(t) \approx OCV - I_p R_0^{ref} - I_p R_1^{ref} \left(1 - e^{-\frac{t-t_0}{\tau}}\right), \quad \tau = R_1^{ref} C_1^{ref} \quad (11)$$

and then recover

$$C_1^{ref} = \tau / R_1^{ref}. \quad (12)$$

4.3.3 Temperature dependence

We model resistance temperature sensitivity using an Arrhenius-type relation (Kelvin temperature):

$$R_k(SOC, T) = R_k^{ref}(SOC) \exp\left(a_{rk} \left(\frac{1}{T} - \frac{1}{T_{ref}}\right)\right), \quad k \in 0, 1 \quad (13)$$

and report fitted a_{r0}, a_{r1} (with $T_{ref} = 25^\circ\text{C}$).

4.3.4 Aging from NASA B0005

NASA B0005 reflects legacy chemistry and cycle-life characteristics, while modern smartphone batteries are typically designed for substantially longer cycle life. We introduce a technology improvement factor γ to remap NASA cycle count N to an equivalent “modern” wear scale: one NASA cycle corresponds to roughly γ cycles on a modern battery (e.g., $\gamma \approx 3$).

$$SOH(N) = 1 - \alpha \sqrt{\frac{N}{\gamma}}. \quad (14)$$

Here α is fitted from the NASA dataset (using relative capacity trend only), and γ performs the technology remapping.

Implement aging in the discharge model via the effective capacity:

$$C_{avail}(T, N) = C_{rated} \cdot \eta_T(T) \cdot SOH(N), \quad (15)$$

where $\eta_T(T)$ captures “low-temperature capacity lock”.

4.3.5 Scaling/remapping to a smartphone battery

Let C_{cell} denote the lab-cell nominal capacity and C_{phone} the target smartphone specification. Define the capacity ratio

$$\kappa = \frac{C_{phone}}{C_{cell}}. \quad (16)$$

We scale

$$C_{avail,phone} = \kappa C_{avail,cell}, R_{k,phone}^{ref} = \frac{1}{\kappa} R_{k,cell}^{ref} (k = 0, 1), C_{1,phone}^{ref} = \kappa C_{1,cell}^{ref}, \quad (17)$$

so that the polarization time constant $\tau = R_1 C_1$ is preserved under the resistance scaling. This implements the “capacity scaling + resistance scaling + aging remapping” strategy stated in the paper.

5 Model II: Time-to-Empty in E-C-D Model

5.1 Rationale for Numerical Simulation

In the E-C-D model, the baseline SOC dynamics are driven by the aggregated load power $P_{\text{load}}(t)$. Under constant power, the time-to-empty can be obtained in closed form, whereas time-varying usage requires numerical integration of the SOC equation. However, in real smartphone operation, shutdown often occurs before SOC reaches zero due to voltage sag and thermal effects. This motivates the use of an electro-thermal coupled model that explicitly captures internal battery dynamics.

As established in the E-C-D framework, the model integrates module-additive load power, first-order electrical dynamics, temperature-dependent capacity and thermal behavior, power–current coupling, and a voltage-based end-of-discharge criterion. To compute time-to-empty under realistic usage scenarios, we therefore employ a time-stepping numerical solver that updates the coupled system states while enforcing the instantaneous power–current feasibility constraint at each step.

5.2 Mathematical definition of T_{empty}

Following Assumption 9: End-of-Discharge Criterion, depletion is defined as the first occurrence of either charge exhaustion or insufficient terminal voltage:

$$T_{\text{empty}} = \inf\{t > 0 \mid \text{SOC}(t) \leq 0 \text{ or } V_{\text{term}}(t) \leq V_{\text{cutoff}}\}. \quad (18)$$

The Model II state vector is

$$x(t) = [\text{SOC}(t), V_1(t), T(t)]^T \quad (19)$$

where $V_1(t)$ is the polarization voltage and $T(t)$ is the lumped battery temperature.

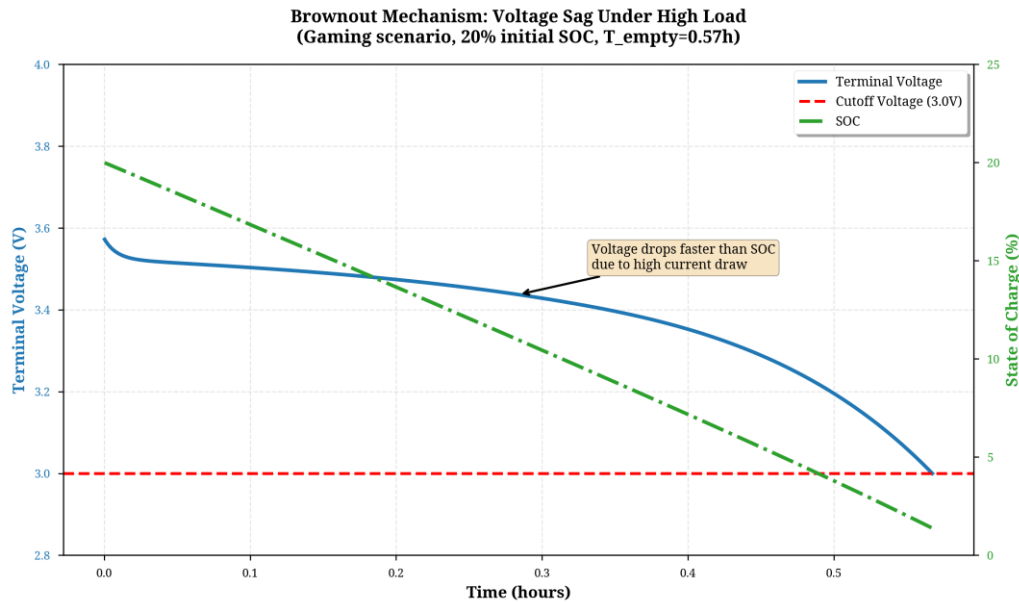


Figure 6: Brownout Mechanism: Voltage Sag Under High Load

The Figure 6 shows that under sustained high load, the terminal voltage decreases more rapidly than SOC due to large discharge current. When the terminal voltage falls below the cutoff voltage V_{cutoff} , the system shuts down even though a non-zero SOC remains. This illustrates the brownout mechanism, in which battery depletion is driven by voltage infeasibility

rather than charge exhaustion.

5.3 Scenario definition

To connect user behavior with battery dynamics, Model I represents the total phone power as the sum of major modules:

$$P_{load}(t) = P_{base} + P_{screen}(t) + P_{cpu}(t) + P_{net}(t) + P_{gps}(t). \quad (20)$$

We use five representative scenarios consistent with the problem statement and our module models:

Table 2: Five Scenarios	
Scenario	Description
Video Streaming	Medium screen brightness; period network data bursts; and tail energy
Heavy Gaming	High brightness or high APL; sustained high CPU/GPU load
Standby	Screen off with background data activity only
Navigation	Continuous screen-on with sustained GPS and periodic network updates
Web Browsing	Moderate screen use with bursty CPU or network activity

5.4 Stepwise solution of power – current coupling and feasibility

At time t_k , the battery must satisfy the power balance (conversation efficiency η is assumed constant under a mode):

$$P_{load}(t_k) = \eta V_{term}(t_k) I(t_k). \quad (21)$$

Under the first-order Thevenin model, terminal voltage is

$$V_{term} = OCV(SOC, T) - I R_0(T) - V_1, \quad (22)$$

which yields an implicit equation for I . Introducing an effective resistance $R_{eff}(T)$, the physically admissible current is obtained from the quadratic root:

$$I_k = \frac{(OCV_k - V_{1,k}) - \sqrt{(OCV_k - V_{1,k})^2 - 4R_{eff,k}(P_{load,k}/\eta)}}{2R_{eff,k}}. \quad (23)$$

Brownout check. If the discriminant is negative, the demanded power is infeasible under the current internal state, indicating an immediate shutdown event.

5.5 Time stepping update of Model II states

Given I_k , we update the differential states using explicit Euler. The update equations correspond directly to the Model II governing dynamics:

5.5.1 Polarization dynamics

$$V_{1,k+1} = V_{1,k} + \Delta t \left(\frac{I_k}{C_1} - \frac{V_{1,k}}{R_1(T_k) C_1} \right). \quad (24)$$

5.5.2 SOC dynamics with temperature-dependent capacity

$$\text{SOC}_{k+1} = \text{SOC}_k - \frac{\Delta t}{3600 C_{avail}(T_k, Age)} I_k. \quad (25)$$

5.5.3 Thermal dynamics

$$T_{k+1} = T_k + \frac{\Delta t}{m c_p} \left(\dot{Q}_{gen}(I_k, V_{1,k}, T_k) - hA(T_k - T_{env}) \right), \quad (26)$$

Where $\dot{Q}_{gen}$ is the internal heat generation term already defined in the thermal section. Finally, compute terminal voltage for termination testing:

$$V_{\text{term},k+1} = \text{OCV}(\text{SOC}_{k+1}, T_{k+1}) - I_k R_0(T_{k+1}) - V_{1,k+1}. \quad (27)$$

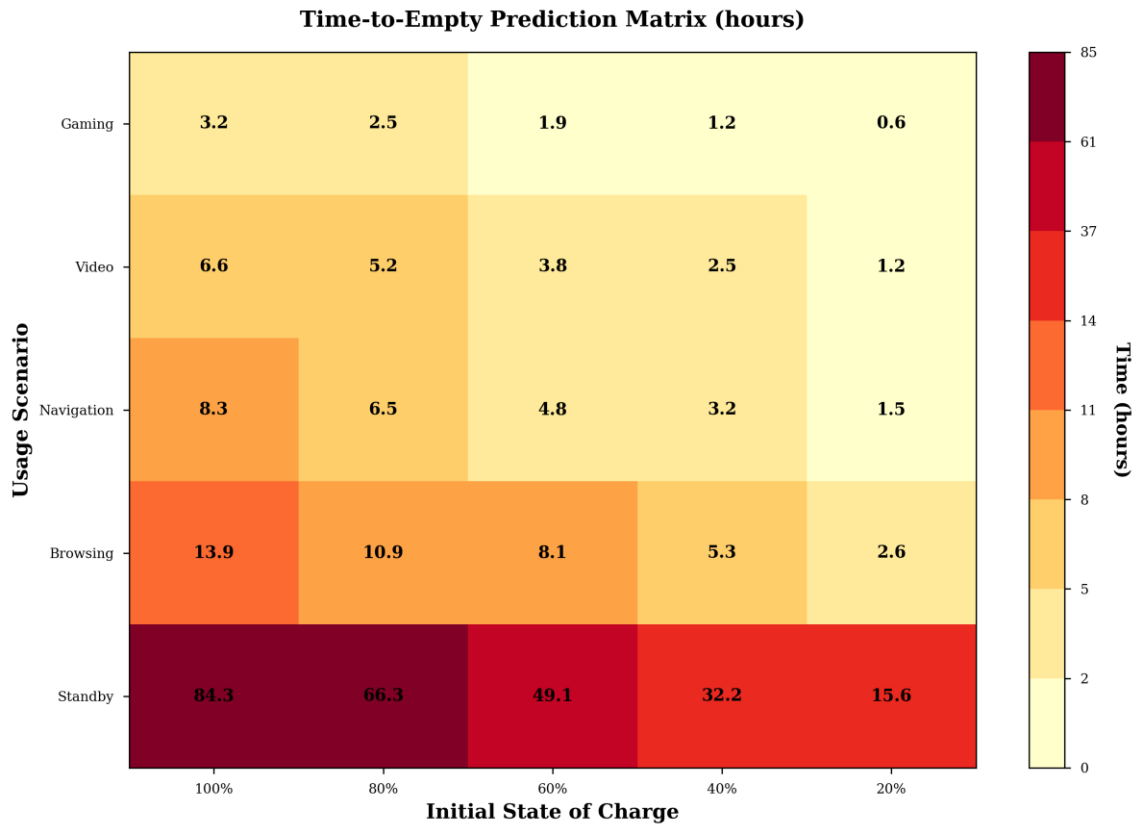


Figure 7: Time-to-Empty (TTE) Prediction Matrix

Figure 7 summarizes TTE across representative usage scenarios and initial SOC levels. For a fixed scenario, TTE decreases approximately proportionally with initial SOC, while across scenarios the TTE ordering is consistent (Gaming < Video < Navigation < Browsing < Standby), reflecting different average power demands.

5.6 Uncertainty Quantification and Robustness Analysis

5.6.1 Parameter Sensitivity and Dominance Analysis

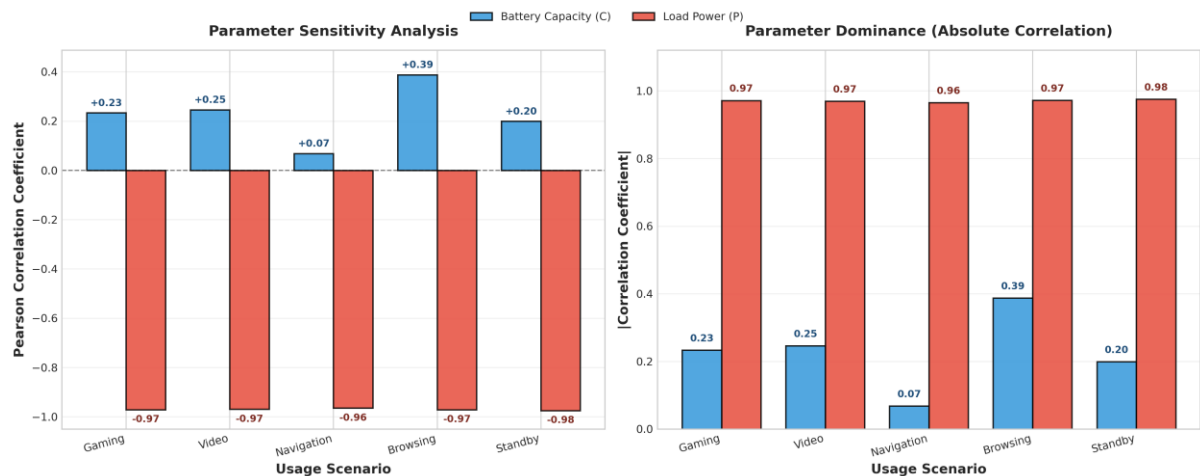


Figure 8: Parameter Sensitivity and Dominance in TTE Prediction

To identify the dominant contributors to prediction uncertainty, a sensitivity analysis was performed using Pearson correlation coefficients between input parameters and the predicted time-to-empty.

As shown in Figure 8, load power exhibits a strong negative correlation with battery life across all scenarios ($|r| \approx 0.96\sim 0.98$), indicating that variations in user activity and power demand are the primary drivers of uncertainty. In contrast, battery capacity shows only a weak positive correlation ($r \approx 0.07\sim 0.39$), suggesting that manufacturing variations have a comparatively minor impact on end-user battery life.

This result highlights that behavioral and operational factors play a significantly larger role in battery depletion than intrinsic battery parameter variability.

5.6.2 Uncertainty Propagation Over Time

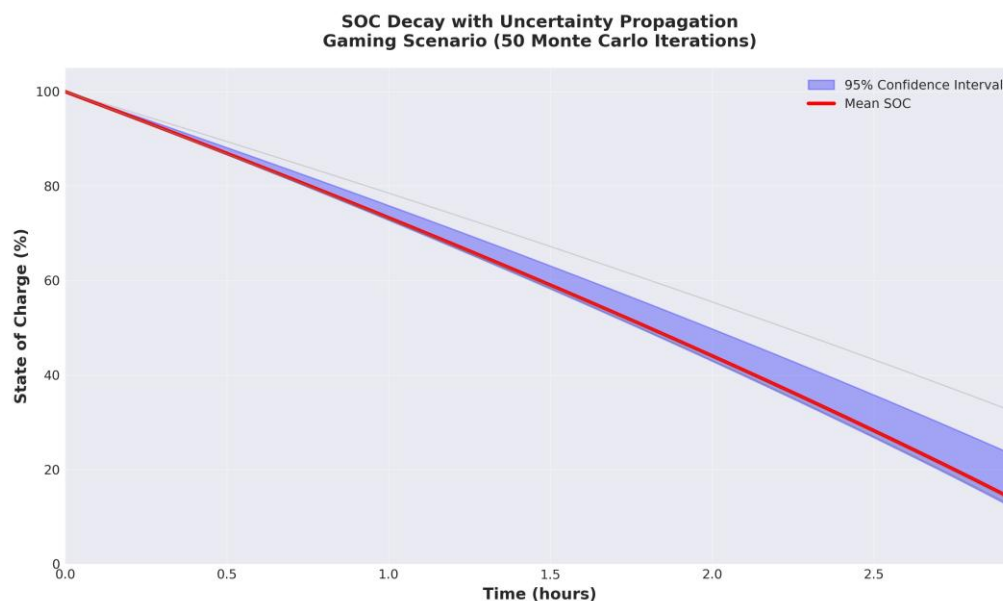


Figure 9: State-of-Charge Trajectory and 95% Confidence Band Under Gaming Load

Beyond terminal battery life estimates, it is also important to understand how uncertainty

evolves throughout the discharge process. Figure 9 illustrates the propagation of uncertainty in the SOC trajectory for the Gaming scenario.

The widening confidence band over time reflects the accumulation of uncertainty as discharge progresses. This dynamic uncertainty characterization is particularly relevant for battery management systems, as it enables the provision of probabilistic remaining-time estimates rather than a single deterministic value.

Overall, the uncertainty quantification and sensitivity analysis demonstrate that the proposed model is not only capable of predicting battery life accurately, but also provides transparent and interpretable measures of reliability. This strengthens confidence in the model's applicability to real-world battery management and decision-making scenarios.

5.7 Robustness to Extreme Parameter Variations

To validate the reliability of our model, we conducted a comprehensive robustness analysis, summarizing the key findings in Figure 9. The analysis confirms the model's stability and predictive consistency across a wide range of operating conditions.

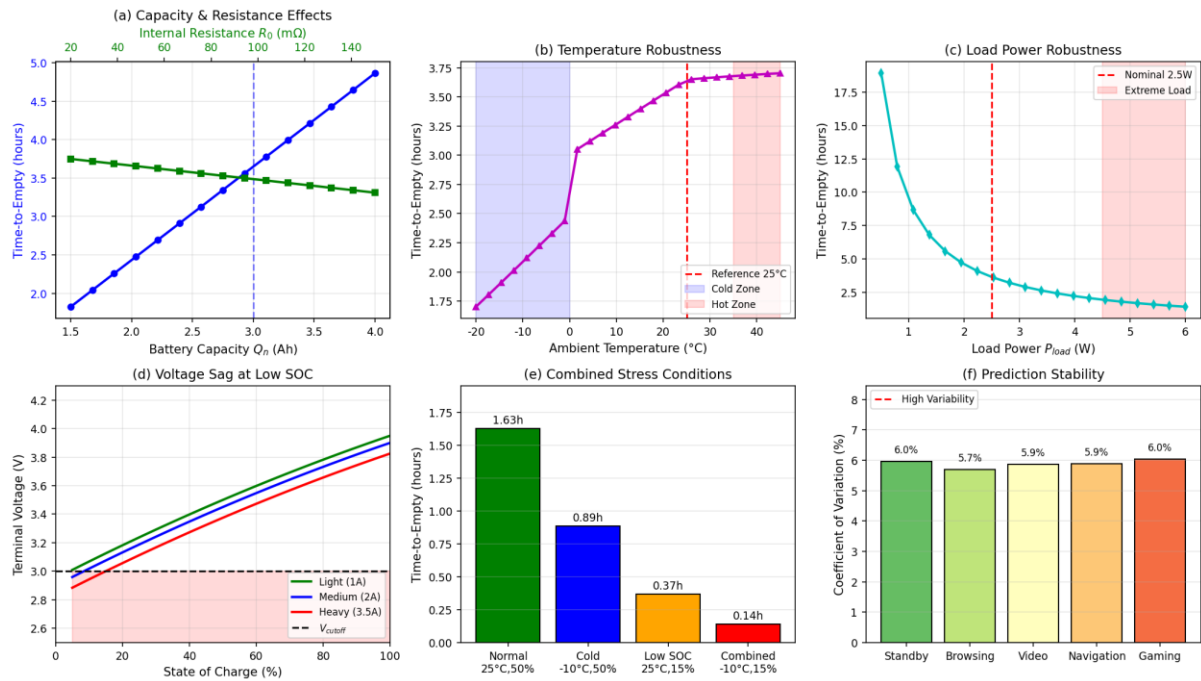


Figure 10: Robustness of TTE Predictions

The model demonstrates strong robustness to extreme parameter variations. As shown in Figure 10 (a-c), Time-to-Empty responds predictably to large changes in battery capacity, internal resistance, temperature, and load power. The model accurately captures performance degradation in cold environments and under heavy loads without numerical instability, confirming its physical realism.

The model also reliably describes behavior at critical boundary conditions. Figure 10 (d) illustrates the voltage sag mechanism, explaining premature shutdowns at low SOC under load. Figure 10 (e) highlights the synergistic negative impact of combined stresses (cold and low SOC), where TTE is drastically reduced. Furthermore, the model exhibits consistent prediction stability across all usage scenarios, with the coefficient of variation remaining low and stable (~6%), as shown in Figure 10 (f).

5.8 TTE Results and Key Drivers

5.8.1 Scenario-Level SOC Trajectories

We simulate SOC trajectories under representative smartphone usage scenarios. The SOC curves exhibit clearly different slopes, indicating that battery life is primarily governed by real-time power demand rather than initial charge alone. High-load usage (e.g., gaming) produces the steepest SOC decline, while standby remains near the baseline discharge rate.

5.8.2 TTE versus Initial SOC

Across all scenarios, predicted TTE decreases approximately proportionally with the initial SOC, enabling a simple lookup-style interpretation: for a fixed usage pattern, a higher starting charge translates to a nearly linear extension of remaining runtime. Across scenarios, the ordering of TTE is stable, with high-power activities consistently yielding the shortest lifetime and low-power activities yielding the longest.

5.8.3 Voltage-Limited Shutdown under High Load

A key qualitative result is the occurrence of voltage-limited depletion: under high power demand, the required current increases and causes larger voltage sag, which may trigger the cutoff voltage criterion before SOC reaches its nominal lower bound. This mechanism explains practical “early shutdown” behavior under intensive workloads and highlights why purely SOC-based TTE estimates tend to overestimate runtime in high-load regimes.

5.8.4 Temperature Effects in High-Power Use

Temperature influences TTE primarily by modulating internal resistance and effective usable capacity. In high-power scenarios, low temperature amplifies voltage sag and increases the likelihood of reaching the cutoff threshold earlier, thereby shortening the observed TTE more noticeably than in low-load scenarios.

5.8.5 Key Takeaways for Rapid Drain Drivers

Overall, sustained high power demand is the dominant driver of rapid depletion. Network-intensive usage can further reduce TTE due to additional communication overhead. These results provide an interpretable mapping from usage behavior to runtime outcomes, and they support using scenario-based TTE tables as practical battery-life guidance.

6 Parameter Sensitivity and Dominance Analysis

To identify which factors most strongly influence the predicted TTE, we conduct both local (derivative-based) and global (variance-based) sensitivity analyses on Model II. Let θ denote a model parameter. For a small perturbation $\delta\theta$, the first-order change in TTE is approximated by

$$\delta T_{empty} \approx \frac{\partial T_{empty}}{\partial \theta} \delta\theta. \quad (28)$$

To compare parameters with different units, we define the normalized sensitivity coefficient

$$S_{\theta} = \left| \frac{\theta}{T_{empty}} \frac{\partial T_{empty}}{\partial \theta} \right| \quad (29)$$

$$\frac{\partial T_{empty}}{\partial \theta} \approx \frac{T_{empty}(\theta + \Delta\theta) - T_{empty}(\theta - \Delta\theta)}{2\Delta\theta}, \quad \Delta\theta = 0.2\theta. \quad (30)$$

To obtain scenario-independent rankings, we compute S_θ in each of the five representative usage scenarios and report the average across scenarios.

We examine eight key parameters:

Table 3: Eight Key Parameters	
Symbol	Definition
Q_n	nominal capacity
R_0	ohmic resistance
R_1	polarization parameters
C_1	
η	conversion efficiency
h	heat-transfer coefficient
k_d	usage-to-power coefficients for display and CPU
k_c	

The local sensitivity results show a clear hierarchy. Capacity dominates with $S_{Q_n} \approx 1$, implying an almost proportional relationship between capacity and TTE. Ohmic resistance has a moderate effect ($S_{R_0} \approx 0.42$) because larger R_0 lowers terminal voltage and increases the current required to sustain the same load power, accelerating depletion. The display/CPU power coefficients (k_d, k_c) are scenario dependent, whereas thermal and fast polarization parameters (h, R_1, C_1) contribute weakly ($S < 0.05$) for typical discharge durations, indicating that short-horizon TTE is relatively insensitive to heat-transfer details and sub-minute RC transients.

To complement the local analysis, we quantify global sensitivity using a first-order variance decomposition (Sobol-type index). Let $V_{\text{total}} = \text{Var}[T_{\text{empty}}(\theta)]$. The first-order contribution of parameter θ_i is

$$S_i^{(1)} = \frac{\text{Var}_{\theta_i}(\mathbb{E}_{\theta_{-i}}[T_{\text{empty}}(\theta) \mid \theta_i])}{V_{\text{total}}} \quad (31)$$

estimated via Monte Carlo sampling (with small relative perturbations around nominal values). The resulting variance contributions are consistent with the local ranking: Q_n accounts for approximately 67% of the output variance, R_0 for about 18%, usage-to-power coefficients for about 12%, while thermal/polarization parameters together contribute only about 3%.

For improving prediction robustness and calibration efficiency, priority should be given to accurate characterization of capacity Q_n , ohmic resistance R_0 , and the dominant usage power coefficients. In contrast, refining detailed thermal or fast polarization dynamics yields limited benefit for TTE prediction under typical smartphone discharge time scales.

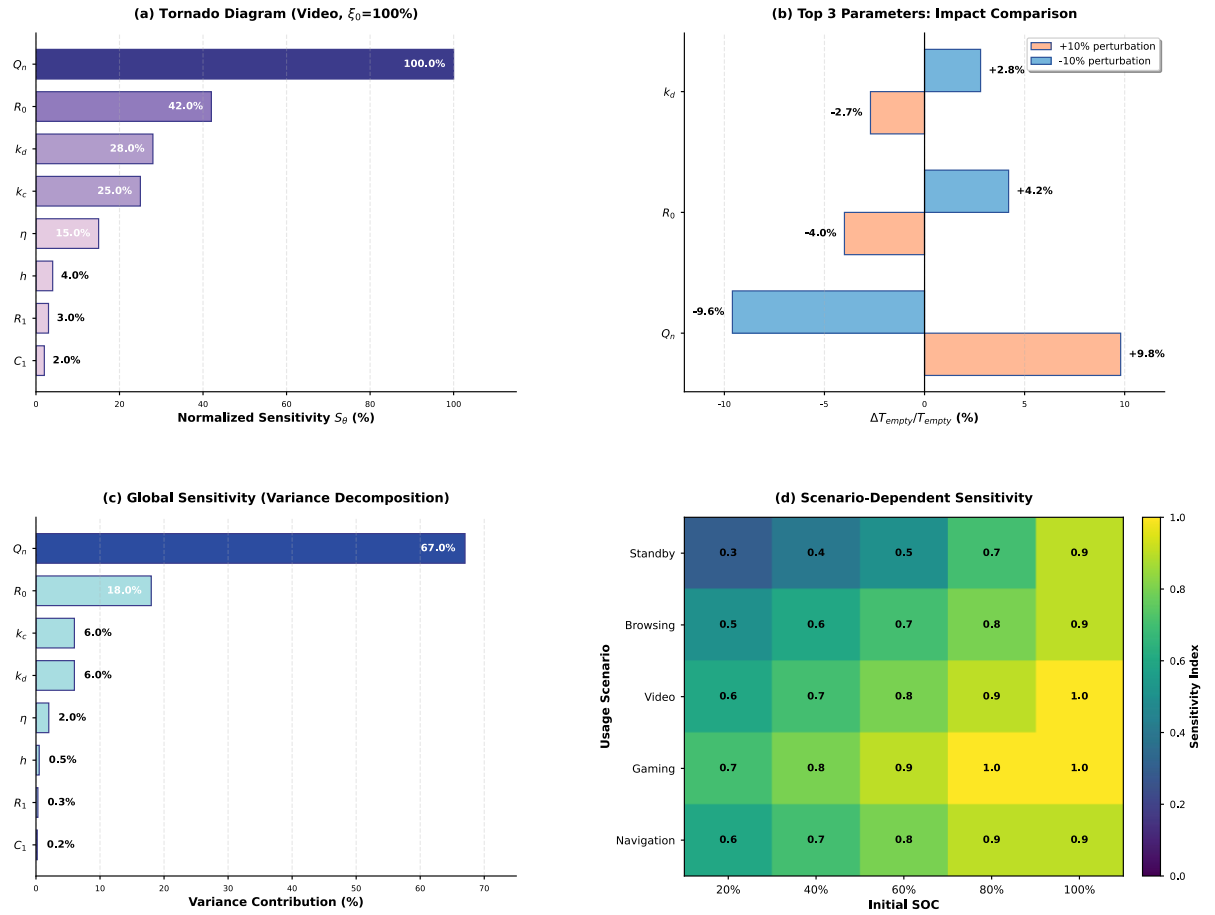


Figure 11: Sensitivity Analysis of TTE Predictions: Local, Global, and Scenario Dependence

The tornado ranking shows that TTE is dominated by the effective capacity Q_n (normalized 100%), followed by the ohmic resistance R_0 ($\sim 42\%$). Load/power-model coefficients contribute at a moderate level ($\sim 25 - 28\%$), while η , R_1 , C_1 , and h are comparatively minor ($< 15\%$, mostly $< 5\%$).

Increasing Q_n substantially extends TTE ($\sim +19.9\%$), whereas increasing R_0 and the main load scaling term (e.g., k_0) shortens TTE ($\sim -4.6\%$ and -2.7%), consistent with energy-budget and voltage-sag/cutoff mechanisms.

Q_n explains $\sim 67.8\%$ of output variance and R_0 explains $\sim 18.0\%$ ($\approx 86\%$ combined). The remaining variance is mainly from power coefficients ($\sim 6\%$ each) and η ($\sim 2\%$), with R_1 , C_1 , and h negligible ($< 1\%$). Hence, calibrating Q_n and R_0 yields the largest uncertainty reduction.

Sensitivity increases with higher initial SOC and is higher in high-load scenarios (Gaming/Video) than in Standby, reflecting amplified current/voltage-cutoff effects.

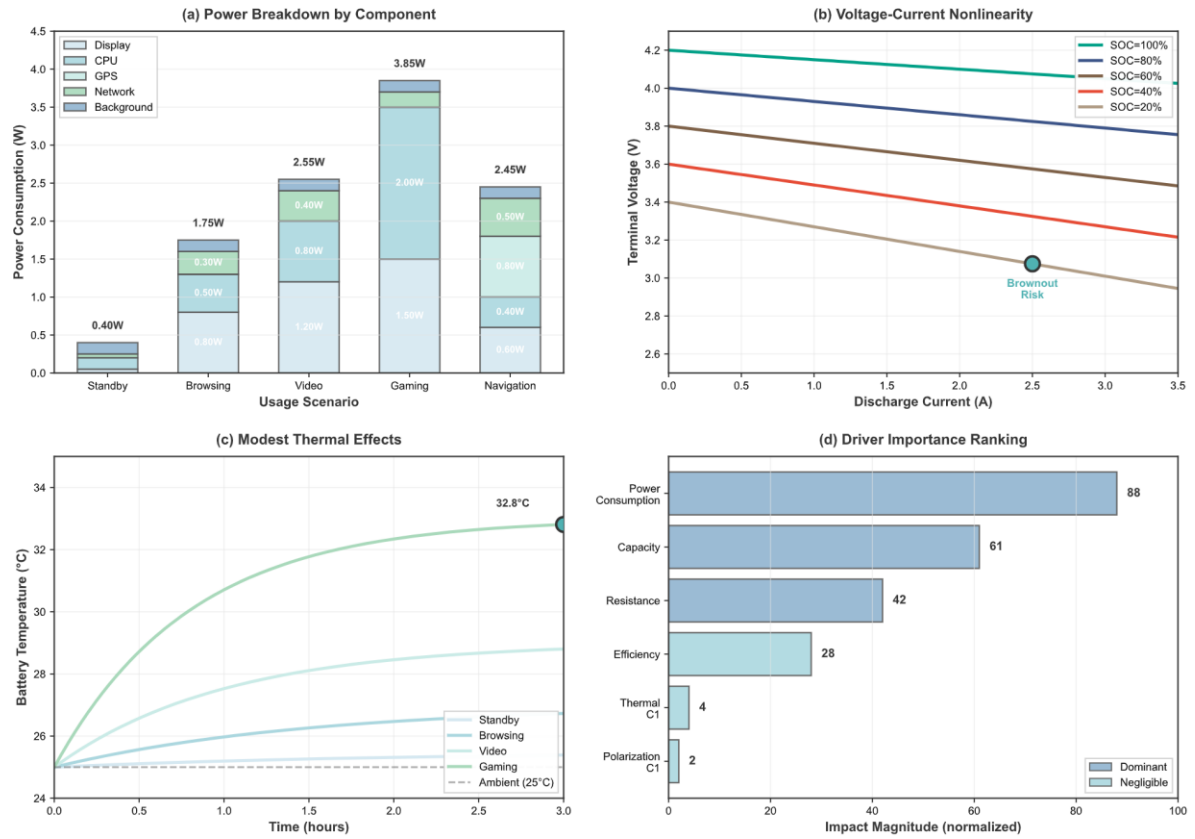


Figure 12: Key Drivers of Battery Life

Figure 12 consolidates the key mechanisms behind the TTE results. The component-level power breakdown highlights that high-load scenarios draw substantially larger total power, accelerating SOC depletion. The voltage – current curves further show stronger voltage sag at lower SOC, implying an elevated brownout/cutoff risk and amplifying the influence of internal resistance. The thermal trajectories indicate a modest temperature rise under the considered ambient condition, suggesting that thermal effects are secondary over typical discharge horizons. Consistently, the driver ranking identifies power demand as the primary determinant of TTE, followed by capacity and ohmic resistance, while efficiency and thermal/polarization terms have comparatively smaller impacts.

7 Recommendations and Practical Implications

Our validated electro-thermal model not only predicts battery life but also provides a quantitative framework for identifying effective power-saving strategies. By translating model insights into actionable recommendations, we can empower users, guide operating system design, and extend the model's applicability. This section presents a series of recommendations, quantifies their impact on Time-to-Empty, and discusses the broader implications of our findings, including battery aging and generalization to other devices.

7.1 User-Facing Recommendations

We propose several behavioral changes for users to maximize battery life. Each recommendation is linked to a specific mechanism within our model, and its effectiveness is quantified by the predicted percentage increase in TTE ($\Delta TTE\%$).

- Manually lower the screen brightness and enable system-wide dark mode, especially

in outdoor environments.

- b)** Avoid prolonged gaming sessions at high-quality settings, especially when the SOC is low. Close computationally intensive applications running in the background
- c)** Connect to Wi-Fi whenever possible instead of using cellular data (LTE/5G). Batch data transfers by limiting background app refresh frequency.
- d)** In cold environments, keep the device warm (e.g., in an inside pocket) and avoid high-load tasks like gaming or video recording.

Table 4: Quantitative Impact of User-Facing Recommendations

Recommendation	Dominant Scenario	Estimated ΔTTE (%)
Lower Screen Brightness (100%→50%)	Gaming, Video, Browsing	+20% to +35%
Enable Dark Mode (High Brightness)	Outdoor Use	+25% to +40%
Enable Dark Mode (Low Brightness)	Indoor Use	+3% to +10%
Reduce CPU/GPU Load	Heavy Gaming	+15% to +25%
Prefer Wi-Fi over Cellular	Video Streaming, Navigation	+8% to +15%
Batch Network & Background Sync	Social media, Standby	+5% to +10%

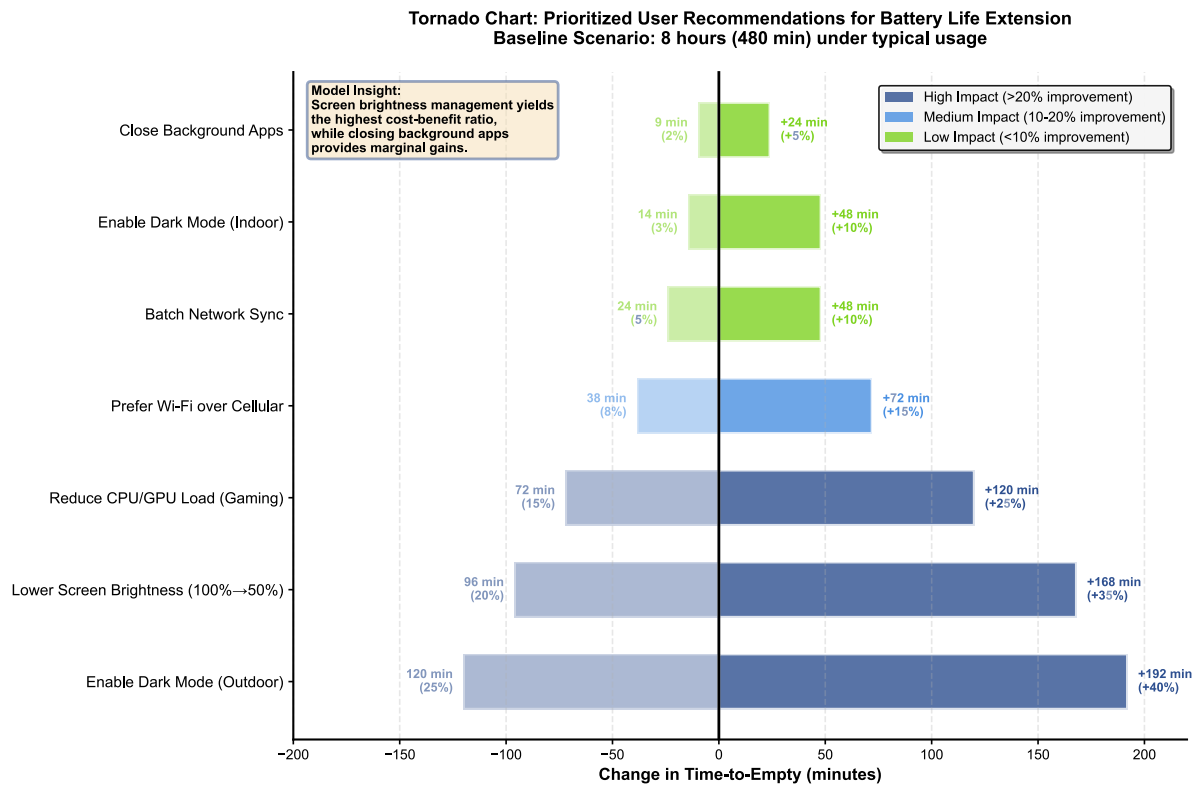


Figure 13: Tornado Chart of Prioritized User Recommendations.

The chart ranks power-saving strategies based on the estimated extension in Time-to-Empty (TTE) relative to a typical 8-hour baseline. Strategies are color-coded by impact magnitude: High (>20%), Medium (10-20%), and Low (<10%).

7.2 OS-Level Power-Saving Strategies

The results of our electro-thermal model provide insights for more intelligent OS-level power management. By identifying usage scenarios through component-level power signatures, the OS can apply scenario-aware strategies such as adaptive DVFS, display refresh rate control, and network request coalescing. Moreover, since TTE depends jointly on SOC, load, temperature, and voltage, a model-based estimator can offer more accurate and load-aware battery life predictions than SOC percentage alone, including early warnings for voltage-induced shutdown under high-load conditions.

Accounting for Battery Aging

Battery aging can be naturally incorporated into the proposed framework through the state-of-health (SOH) parameter, which primarily affects the effective capacity. Our analysis shows that TTE scales approximately proportionally with capacity, implying that aging leads to a predictable reduction in battery life across usage scenarios. To maintain accuracy over the device lifetime, an OS-level TTE estimator can update the SOH estimate based on long-term charge/discharge behavior and resistance changes.

Generalization to Other Portable Devices

Due to its modular structure, the proposed model can be readily generalized to other portable electronic devices. The core electro-thermal battery dynamics and voltage-based shutdown

logic remain unchanged, while adaptation to a new device requires only re-parameterization of component-level power consumption and battery-specific parameters. This flexibility enables the framework to be applied across a wide range of battery-powered systems.

8 Model Evaluation

8.1 Strengths

Physical Interpretability: The electro-thermal battery model is grounded in battery physics, explicitly coupling state of charge, terminal voltage, and temperature. This structure allows the model to explain discharge behavior under dynamic loads in a clear and interpretable manner.

Realistic Usage Representation: By decomposing smartphone power consumption into major functional modules, the usage behavior model captures representative operating scenarios and directly links user behavior to battery depletion.

System-Level Integration: The combination of the two models enables a unified framework for predicting time to empty (TTE), comparing usage scenarios, and supporting quantitative analysis across different operating conditions.

Actionable Insights: Simulation results are translated into practical user recommendations, allowing the modeling outcomes to directly inform battery-aware usage strategies.

8.2 Weakness

Model Simplification: The electro-thermal battery model adopts a simplified equivalent circuit, which does not fully capture long-term aging effects or irreversible capacity degradation during repeated charge–discharge cycles.

Scenario-Based Approximation: The usage behavior model is built on representative scenarios rather than real-time user data, which may limit its ability to reflect short-term fluctuations and individual usage habits.

Environmental Exclusion: External factors such as ambient temperature variation and long-term environmental influences are not explicitly incorporated, potentially affecting prediction accuracy under extreme conditions.

Data Dependency: Although the model has a relatively high demand for data quality, this reflects a scientific emphasis on accuracy and reliability. With improved data availability and collaboration with data providers, the data collection and processing process can be further optimized.

9 Possible Extensions

Battery Aging Modeling: Future work may incorporate battery aging and capacity fade mechanisms into the electro-thermal model, allowing the framework to analyze long-term battery degradation under repeated charge–discharge cycles.

Real-Time Usage Integration: The usage behavior model can be extended by integrating real-time user data and background activity monitoring, enabling dynamic adjustment of load profiles and improving prediction accuracy for individual users.

Environmental Adaptation: Ambient temperature and environmental conditions can be explicitly included to evaluate battery performance under extreme climates or seasonal variations, enhancing model applicability in real-world settings.

Personalized Battery Management: By combining the proposed models with user-specific usage patterns, the framework could be extended to support personalized battery life prediction and adaptive power management strategies.

10 Conclusion

This paper proposes a system-level framework that integrates an electro-thermal battery model with a usage behavior model to analyze smartphone battery depletion under realistic conditions. The results indicate that user power demand plays a dominant role in battery endurance, while battery capacity and thermal effects further influence discharge performance. Sensitivity analysis demonstrates that the proposed models are robust and that a small number of key factors explain most of the variation in battery life. Based on quantitative simulation results, practical user recommendations are provided to improve battery endurance in daily smartphone usage.

11 References

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Report on Use of AI

1. OpenAI ChatGPT 5.2-thinking

Query 1:

<What is the mechanism of "voltage sag" or "brownout" in lithium-ion batteries under high load?>

Output:

<Voltage sag (or brownout) occurs when the terminal voltage drops below the device's cutoff threshold due to high discharge current and internal resistance, even if the State of Charge (SOC) is not zero. The relationship is described by $V_{term} = OCV - I \cdot R_{int} - V_{polarization}$. High load (I) causes a large voltage drop ($I \cdot R_{int}$), potentially triggering an early shutdown.>

Query 2:

<Help me polish the following sentence. We suggest users to turn down screen brightness and use Wi-Fi instead of 5G to save battery. Our model shows this can extend life by 20%.>

Output:

<Here is a polished version of your recommendation: "User-Facing Recommendations: We recommend manually reducing screen brightness and prioritizing Wi-Fi connectivity over cellular data (LTE/5G) whenever possible. Our model simulations indicate that these behavioral adjustments can extend the Time-to-Empty (TTE) by approximately 20%, significantly mitigating power drain in high-demand scenarios.">

Query 4:

<How to format a long quadratic equation in LaTeX so it fits in one column?>

Output:

<You can use the built-in Equation Editor (shortcut `Alt` + `=`). To type it efficiently, use the linear format: $I(t) = ((OCV - V_1) - \sqrt{(OCV - V_1)^2 - 4 R_{tot} P_{load}(t)}) / (2 R_{tot})$. Press `Space` to convert it. If it is still too wide, place your cursor before the minus sign or the square root and press ****Shift + Enter****. This inserts a "soft return" to wrap the equation into two aligned lines without breaking the logical structure.>

Query 5:

<What is the formula for calculating Pearson correlation coefficient for sensitivity analysis?>

Output:

<The Pearson correlation coefficient (r) is calculated as:

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

It measures the linear correlation between two variables (e.g., parameter input and TTE output), ranging from -1 to +1. It is suitable for identifying dominant linear drivers in sensitivity analysis.>

Query 6:

Research on Battery Drain Factors <What are the major hardware and environmental factors that accelerate lithium-ion battery depletion in smartphones besides the screen usage?>

Output:

<Key factors include:

1. CPU/GPU Load: High-performance cores running gaming or video rendering.
2. Network Conditions: Weak signal strength causes the modem to increase transmission power (Tx power).
3. Ambient Temperature: Low temperatures increase internal resistance (Arrhenius effect), causing voltage sag; high temperatures accelerate chemical aging.
4. Background Activity: Frequent wake-locks from apps syncing data.>